

Name: Answers

Date: _____

Class: _____

**Year 8 Mathematics****Mini-Test 5, 2018****Topic – Circles and Algebra**

/ 39

%

Total Time: 35 min**Equipment:** Calculator and 1 page of notes A4 (one sided). *SHOW ALL WORKING OUT*

A Grade	
Determines the radius and diameter of circles, given their circumference or area and the perimeter and area of composite figures involving circles, quadrants or semicircles.	
Solves a variety of problems involving circles and part circles, giving approximate or exact answers as required.	
Flexibly moves between simplified and complex algebraic expressions, choosing the most appropriate form where variables are to the power of one.	
B Grade	
Determines the circumference, perimeter and area of circles and simple composite figures involving circles, quadrants and semicircles.	
Plans and carries out the solution to practical problems involving circles, quadrants or semicircles, including those which involve more than one step.	
Moves between simplified algebraic expressions where variables are to the power of one.	
C Grade	
Chooses appropriate formulas to calculate the circumference, in terms of diameter or radius, and area of circles in routine or familiar problems.	
Solves simple and familiar practical problems involving circles.	
Moves between factorised and expanded algebraic expressions involving a single numerical common factor.	
D Grade	
Uses formulas to calculate the circumference and area of circles, but sometimes confuses radius and diameter or circumference and area.	
Attempts solutions to some practical problems involving circles, but confuses situations where circumference or area are involved.	
Uses a model to simplify an algebraic expression	
E Grade	
Does not meet the requirements of a D grade.	

Teacher Feedback:

Question 1

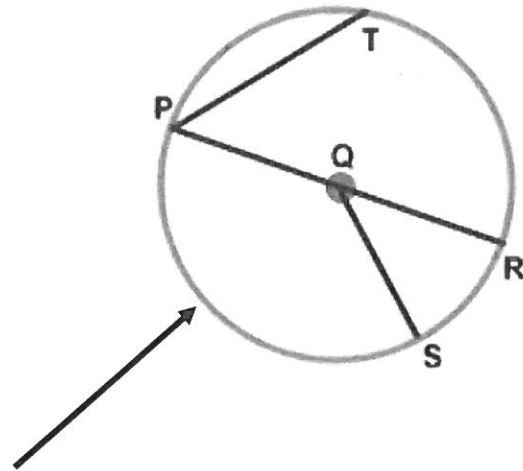
Which of the following is the diameter?

- a) PT
- b) PQ
- c) PR
- d) SQ

Question 2

Which of the following is the radius?

- a) PT
- b) PQ
- c) PR
- d) All of the above

**Question 3**

The arrow is pointing to the:

- a) Center of the circle
- b) Circumference
- c) Diameter
- d) Pi

Question 4

Pi is an irrational number that is derived from:

- a) Circumference $\div$ diameter
- b) Diameter $\div$ radius
- c) Area $\div$ circumference
- d) Circumference $\times$ radius

Question 5

A like term for $8abc$ is:

- e) $3cab$
- f) $8ab$
- g) $8ac$
- h) $8bcc$

Question 6

Which of the following is obtained when $3p+5+6p-2$?

- a) $12p$
- b) $14p+2$
- c) $9p-3$
- d) $9p+3$

Question 7

In the algebraic expression $3x+7$, which is the coefficient?

a) $3x$

b) x

c) 3

d) 7

Question 8

In the algebraic expression $18ab-8b+2$, which is the constant?

a) 2

b) $18ab$

c) a

d) -8

Question 9

Which is the most simplified version of $3y+y+y+y$?

e) $3y+3y$

f) $3y^3$

g) $9y^3$

h) $6y$

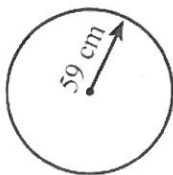
Part B – Short Answer ** Answer Each Question and Show Full working ** 30 marks

Question 10

[10 marks]

Find the circumference/perimeter of the following circles/shapes:

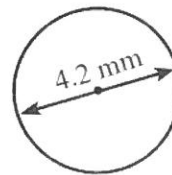
i)



$$2 \times \pi \times 59$$

$$= 370.71 \text{ cm}$$

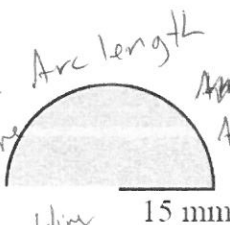
ii)



$$\pi \times 4.2$$

$$= 13.19 \text{ mm}$$

iii)



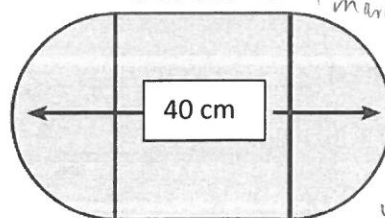
$$\text{Arc} = \frac{1}{2} \times 2 \times \pi \times 15$$

$$= 47.12 \text{ mm}$$

$$\text{Arc} + \text{Diameter} = 47.12 \text{ mm} + 30 \text{ mm}$$

$$= 77.12 \text{ mm}$$

iv)



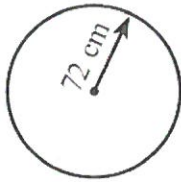
$$\begin{aligned} p &= 2 \times 10 \times \pi + 20 + 20 \\ &= 62.88 + 40 \\ &= 102.88 \text{ cm} \end{aligned}$$

Question 11

[10 marks]

Find the area of each of the following shapes:

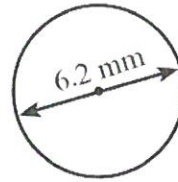
i)



$$A = 72^2 \times \pi \checkmark$$

$$= 16286.02 \text{ cm}^2 \checkmark$$

ii)

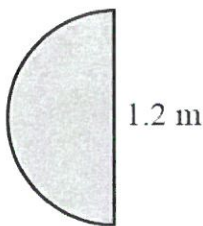


$$A = 3.1^2 \times \pi \checkmark$$

$$= 30.19 \text{ mm}^2 \checkmark$$

-1 mark total for wrong units

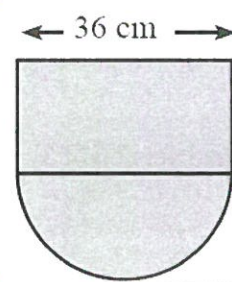
iii)



$$\frac{1}{2} \times \pi \times 0.6^2 \checkmark$$

$$= 0.57 \text{ m}^2 \checkmark$$

iv)



$$36 \times 14 \checkmark$$

$$= 504 \text{ cm}^2$$

$$\frac{1}{2} \times \pi \times 18^2$$

$$= 508.94$$

$$504 + 508.94 = 1012.94 \text{ cm}^2$$

1 mark for correct radius
1 mark for halving
1 mark for correct answer

1 mark for area of rectangle
1 mark for Area of $\frac{1}{2}$ circle
1 mark for adding [4 marks]

Question 12

Arthur ran around a circular field 3 times. If he ran a total distance of 750 m, what is the diameter and radius of the field? (round your answer to 2 d.p.)

$$750 \div 3 = 250 \text{ m circumference} \checkmark$$

$$250 \div \pi = 79.58 \text{ m diameter} \checkmark$$

$$79.57 \div 2 = 39.79 \text{ m radius} \checkmark$$

+1 mark for correct rounding

Question 12

[6 marks]

Simplify each expression where possible by collecting like terms.

(a) $23y - y$ $22y$

(b) $10ab - 7b$ $10ab - 7b$

(c) $2a^2 + 5 + 6a^2$ $8a^2 + 5$

(d) $2x - 5x + 4$ $-3x + 4$

(e) $2p + 9q + 3p - 8q$ $5p + q$

(f) $3m + 10 + 4n + m - 6$ $4m + 4n + 4$